

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) — Semester VI Examination, 2022-23

Paper No. MTB-601: Set Theory and Metric Spaces

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) Let $X = \mathbb{R}$ be the set of all real numbers. For $x, y \in X$, define $d(x, y) = |x^2 - y^2|$. Show that d is not a metric on X . [2]
- (b) Define discrete metric space. Prove that in the discrete metric space (X, d) , every subset $G \subset X$ is an open set. [2]
- (c) Give an example of a pseudometric which is not a metric. [2]
- (d) Let A be a subset of the metric space (X, d) . Prove that $\text{diam}(A) = 0$ if and only if A contains at most one point. Here $\text{diam}(A)$ denotes the diameter of A . [2]
- (e) Let (X, d) be a metric space. Prove that every finite set in X is closed. [2]
- (f) Show that the set of all rational numbers $\mathbb{Q}$ is not open in the metric space $\mathbb{R}$ with the usual metric. [2]
- (g) By giving an example, show that $\overline{A \cap B} \neq \overline{A} \cap \overline{B}$, where A, B are two subsets of the metric space (X, d) and $\overline{A}$ denotes the closure of A . [2]
- (h) In a metric space (X, d) , prove that a Cauchy sequence is bounded. [2]
- (i) Prove that $\mathbb{Z}$, the set of integers, is not dense in $\mathbb{R}$. [2]
- (j) Prove that $\text{card}[0, 1] = \text{card}(0, 1]$. [2]
- (k) Define partial order relation and give an example of a partial order relation. [2]

Question 2.

- (a) Let W be the space of all sequences in $\mathbb{R}$. Define

$$d(x, y) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{|x_n - y_n|}{1 + |x_n - y_n|}$$

where $x = (x_n)$ and $y = (y_n)$ are in W . Prove that (W, d) is a metric space. [6]

- (b) Let (X, d) be a metric space and $Y \subset X$. Prove that a set A is open in (Y, d_Y) if and only if there exists a set G open in (X, d) such that $A = G \cap Y$. [6]

Question 3.

- (a) Let (X, d) be a metric space. Prove that every closed sphere in X is a closed set. [5]
 (b) Let (X, d) be a metric space and let the metric d^* on X be defined by

$$d^*(x, y) = \frac{d(x, y)}{1 + d(x, y)}, \quad \text{for } x, y \in X.$$

Prove that d^* is equivalent to d . [7]

Question 4. Let A be a subset of a metric space (X, d) . Prove that: [3 each]

- (a) $X - A^\circ = \overline{X - A}$, where A° denotes the interior of A .
 (b) $\text{diam}(\overline{A}) = \text{diam}(A)$, where $\text{diam}(A)$ denotes the diameter of A .
 (c) A is closed if and only if A contains all its limit points.
 (d) Define boundary ∂A of A . Prove that $\partial A = \overline{A} \setminus A^\circ$.

Question 5.

- (a) Prove that the space $C[0, 1]$ of all continuous real-valued functions defined on $[0, 1]$ with the metric d defined by

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in [0, 1]\}$$

is a complete metric space. [6]

- (b) Let (X, d) and (Y, ρ) be any two metric spaces. A function $f : X \rightarrow Y$ is continuous on X if and only if for each open set $G \subset Y$, $f^{-1}(G)$ is an open subset of X . [6]

Question 6.

- (a) Let (X, d) and (Y, ρ) be two metric spaces and $f : X \rightarrow Y$ be a bijective isometry. Then prove that f is continuous. [5]
 (b) Let (X, d) and (Y, ρ) be two metric spaces and $f : X \rightarrow Y$ be a uniformly continuous function. Let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in X . Prove that $(f(x_n))_{n=1}^\infty$ is a Cauchy sequence in Y . [5]
 (c) Define homeomorphism. Give an example of a homeomorphism. [2]

Question 7.

- (a) Define 'countable set' and 'uncountable set'. Give one example of each. Prove that $(0, 1)$ is uncountable. [6]
- (b) Prove that a countable union of countable sets is countable. [6]